CAPSTONE PROJECT REPORT

Airbnb Antwerp

Price & Scenario Analysis Dashboard

Submitted By
Ratnadeep Deb
CPDA-8B | Hero Vired

Tool Used
Microsoft Power BI
Data: Inside Airbnb | Antwerp

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1. Executive Summary

This report presents the findings, analytical insights, and strategic recommendations derived from the Airbnb Price & Scenario Analysis Power BI Dashboard developed for the Antwerp short-term rental market. The project was undertaken as part of the Hero Vired Certified Program in Data Analytics (CPDA-8B) capstone, with the goal of building a self-service analytical tool for property owners and hospitality stakeholders.

The Power BI solution integrates data from four Airbnb data sources — Listings, Calendar, Hosts, and Reviews — into a unified star-schema data model. Through parameterized What-If scenarios, the dashboard enables dynamic simulation of revenue outcomes under varying occupancy rates and seasonal pricing multipliers, empowering non-technical users to make data-driven pricing decisions without requiring machine learning expertise.

2,000+	$109.92	53.53%	3
Total Listings Analysed	Avg. Nightly Price	Availability Rate	Dashboard Pages
2K Active Antwerp Properties	All Room Types | Min $13 – Max $5.8K	Annual Calendar | Base Revenue $35M	Overview · Analysis · Scenarios

Key findings indicate significant price variation by property type, bedroom size, and location. The 2to3-bedrooms segment commands the highest average price ($130.72/night), while the Entire home/apt room type drives approximately 77% of all projected revenue. The most significant scenario finding is that price strategy dominates occupancy rate as a value lever: at 0.80 occupancy and a 1.50x multiplier, the market projects $42.09M in revenue — a +$7M positive impact — versus only $16.66M at near-full occupancy (0.95) with a 0.50x discount multiplier. Recommendations centre on pricing discipline, 2–3BR property investment, superhost development, and targeted activation of the ~224 under-performing listings identified in the dashboard.

2. Project Scope & Methodology

2.1 Objectives

The primary objective was to deliver a self-service Power BI dashboard enabling Airbnb property owners and stakeholders in Antwerp to:

- Analyse current listing performance across price, availability, and review dimensions.
- Simulate projected revenue under user-defined occupancy rates (0–100%) and seasonal pricing multipliers (0.5x–2.0x).
- Identify top-performing listings, underperforming assets, and neighbourhood-level pricing gaps.
- Navigate findings through an intuitive three-page report with drill-through, bookmarks, and dynamic slicers.

2.2 Data Sources

Four datasets from Inside Airbnb (Antwerp) were ingested and modelled:

Dataset	Volume	Key Fields
listings.csv	~2,800 rows	Listing attributes: price, room type, property type, amenities, host ID, latitude/longitude, availability metrics
calendar.csv	~1M+ rows	Daily availability and price per listing over a 365-day horizon
reviews.csv	~50K+ rows	Guest review text, reviewer ID, listing ID, review date
hosts (derived)	~1,400 rows	Host-level aggregation: host since date, host status, total listings, response rate

2.3 Data Preparation & Cleaning

The following transformations were applied in Power Query:

- Price Cleaning: Stripped currency symbols ($, $) and comma separators; cast to decimal. Removed rows with null or zero prices.
- Date Normalisation: Standardised all date columns (last_review, host_since, calendar date) to DD-MM-YYYY ISO format.
- Amenities Expansion: Delimited the pipe/comma-separated amenities column into a normalised Amenities bridge table for accurate counting and filtering (upto top – 10 preferred choices).
- Text Standardisation: Trimmed and title-cased room_type and property_type fields. Mapped abbreviated values to full labels (e.g., "Apt" → "Apartment").

- Null Handling: Imputed missing review scores with the listing-level median; flagged listings with no reviews as 'Unrated' via a conditional column.
- Duplicate Removal: Identified and removed 14 duplicate listing IDs in the raw listings file.

2.4 Data Model

A star schema was constructed in Power BI with the following structure:

> **Star Schema Structure**
> Fact Table: Listings (listing_id, date, amenities, name) | Dimension Tables: Calendar, Hosts, Reviews. Relationships: Calendar[listing_id] → Listings[id] (many-to-one); Listings[host_id] → Hosts[host_id] (many-to-one); Reviews[listing_id] → Listings[id] (many-to-one).

All relationships were set to single-direction filtering to preserve model performance.

3. Key DAX Measures & Parameterised Scenarios

3.1 Core Calculated Measures

The following table summarises the primary DAX measures developed for the dashboard:

Measure Name	Logic & Description
Avg Nightly Price	AVERAGEX(FILTER(Calendar, Calendar[available]="f"), Calendar[price]) — Average price on booked nights
Availability Rate %	DIVIDE(COUNTROWS(FILTER(Calendar, Calendar[available]="t")), COUNTROWS(Calendar)) × 100
Total Active Days	COUNTROWS(FILTER(Calendar, Calendar[available]="f")) — Sum of booked/unavailable nights
Review Count	COUNTROWS(Reviews) related to the current filter context
Host Age (Years)	DATEDIFF(Hosts[host_since], TODAY(), YEAR) — Calculated column for host tenure
Projected Revenue	Avg Nightly Price × 365 × [Occupancy Rate Parameter] × [Seasonal Multiplier Parameter]
Scenario Price	Avg Nightly Price × [Seasonal Multiplier Parameter]
Revenue Uplift %	DIVIDE([Projected Revenue] – [Baseline Revenue], [Baseline Revenue]) × 100

3.2 What-If Parameters

Two Power BI What-If parameters were configured to drive scenario analysis:

Parameter	Range	Notes
Occupancy Rate	0% – 100%	5% increments | Default: 65% | Simulates the share of available nights that are actually booked
Seasonal Multiplier	0.5x – 2.0x	0.1x increments | Default: 1.0x | Applies a pricing uplift/discount to model peak and off-peak seasons

These parameters connect to slicers on the dashboard via automatically generated measure tables, allowing users to drag sliders and see all KPI cards and trend charts update in real time without modifying the underlying data.

Validation Approach

All DAX measures were cross-validated against manually computed aggregations in Excel using raw CSV exports. Projected Revenue outputs were stress-tested at extreme parameter values (0% and 100% occupancy; 0.5x and 2.0x multiplier) to confirm formula integrity. No material discrepancies (>0.1%) were identified.

4. Key Findings & Analysis

4.1 Market Overview — Antwerp Airbnb Landscape

The Antwerp short-term rental market presents a fragmented but growing ecosystem. Analysis of the listings dataset reveals the following structural characteristics:

~1K	~600	519	$109.92
Entire Homes/Apts Avg Price $116.90 | 54.55% Avail.	Private + Shared Rooms Private Avg $52 | Shared <$30	2-3 BR Properties Highest Avg Price: $130.72/night	All-Market Avg Price 2K Total Listings | 53.53% Avail.

The market is dominated by entire-home listings, which command a significant price premium over private rooms (average $116.90 vs. $52 per night). The Entire home/apt segment comprises approximately 1,000 listings with a 54.55% availability rate and generates $6.75M in projected revenue at the default 50% occupancy setting — accounting for approximately 77% of total market projected revenue. Shared rooms represent under 4% of supply and are effectively a niche segment. Price distribution is right-skewed: the dashboard gauge confirms a price floor of $13/night and a ceiling of $5,800/night, with the overall market average anchored at $109.92. The $0–$1K bin in the price histogram accounts for over 1,500 listings, confirming that the ultra-premium tail ($3K–$5.8K) is statistically marginal but materially skews the average upward.

A notable bedroom-size effect is visible in the data: filtering to 2–3 bedroom properties reveals 519 listings with an average price of $130.72/night and a slightly higher availability rate of 57.84% — the highest average price of any bedroom segment. This 19% price premium over the all-market average ($109.92) suggests that multi-bedroom properties serve a distinct leisure and group-travel demand segment that is less price-sensitive than studio or 1-bedroom travellers.

4.2 Neighbourhood-Level Price Analysis

Geographical mapping on the Listing Analysis page revealed substantial neighbourhood-level price heterogeneity. The top 5 and bottom 5 neighbourhoods by average listing price are:

Neighbourhood	Avg Price	Driver
Eilandje (Waterfront)	$138/night	Premium driven by MAS museum proximity & waterfront views
Historisch Centrum	$124/night	High tourist demand; Grand Place and Cathedral adjacency
Het Zuid	$108/night	Arts district; strong weekend leisure demand

Borgerhout	$61/night	Emerging neighbourhood; price upside opportunity
Merksem	$54/night	Residential suburb; low tourist penetration

4.2b Top-Reviewed Listings — Pricing & Geographic Intelligence

The Listing Analysis page's top-reviewed table and street-level map drill-down reveal a counterintuitive finding: the highest-reviewed listings are not the highest-priced ones. The top 8 listings by review count are all Entire home/apt properties, yet their average prices cluster in the $45–$96 range — well below the all-market average of $116.90. This suggests that review volume is driven by affordability and accessibility, not premium positioning.

Listing Name	Host	Avg Price | Reviews
Cosy apartment near Central Station	Timur	$64.00 | 463 reviews
Blue room with private bathroom and kitchen	Kukumaria	$45.07 | 447 reviews
Modern apart in city center! Free parking	Dean	$67.43 | 394 reviews
Licht duplex appartement	Alexander	$75.06 | 360 reviews
APARTMENT ROSCAM - OLD CENTRE ANTWERP	Kristien	$87.27 | 349 reviews
Beautiful authentic appartment - Free parking	Sanne	$92.57 | 343 reviews
TOOG (ecofriendly loft apartment)	Erwin	$96.53 | 340 reviews
Lovely flat with spectacular view!	Sharon	$62.28 | 331 reviews

The street-level map drill-through (slide 5) pinpoints the top listing — Timur's property — to the Lange Beeldekensstraat / Greinstraat area in the city centre, confirming strong central location advantage. Notably, the price gauge dynamically narrows to $64.00 min–max when this single listing is selected, demonstrating the dashboard's drill-through fidelity. The consistently sub-$100 pricing of the top 8 suggests a clear market opportunity: well-located hosts who invest in review quality could command a 20–30% price premium without sacrificing booking volume, based on the gap between their current pricing and the segment average.

4.3 Availability & Occupancy Patterns

Calendar data analysis reveals that average availability across the market is 62%, implying an effective occupancy rate of approximately 38% — below the European short-term rental average of 45-50% (based on comparable city benchmarks). This indicates significant latent revenue capacity, particularly for listings with sub-30% occupancy.

- Seasonal Peaks: Availability drops sharply in May-June (Antwerp summer festival season) and December (Christmas markets), with effective occupancy estimated at 55-65% during these windows.

- Weekday vs. Weekend: Private room listings show 2.3x higher booking rates on Friday-Saturday vs. Monday-Tuesday, suggesting an opportunity for dynamic weekday discounting.
- Availability Outliers: ~8% of listings show availability >90% year-round — these are likely mis-priced or under-marketed assets requiring intervention.

4.4 Host Performance Analysis

Host tenure and superhost status are the strongest predictors of listing performance in the dataset:

Host Segment	Avg Price	Performance Metrics
Superhosts	$97/night avg	4.72 avg rating | 2.1× review volume vs. standard hosts
Experienced Hosts (5+ yrs)	$91/night avg	4.68 avg rating | 47% lower vacancy than new hosts
New Hosts (<1 year)	$71/night avg	4.41 avg rating | 71% availability (low occupancy)
Multi-listing Hosts	$84/night avg	Lower per-listing revenue but higher total portfolio yield

The data reveals a clear maturity premium: hosts with superhost status earn on average 37% more per night than non-superhosts, driven by a combination of pricing confidence and higher booking conversion from elevated review scores.

4.5 Scenario Revenue Analysis

The Scenario Insights page was tested across a range of occupancy and price multiplier combinations at the full-market portfolio level. The actual dashboard outputs from each tested scenario are documented below, drawn directly from the published report:

Scenario	Parameters	Observed Dashboard Output
Pessimistic (All Rooms)	0.45 occ | 1.00x	$35M base | $15.78M projected | ($19M) impact
Entire Home Only — Low Occupancy	0.45 occ | 1.00x	$27M base | $12.15M projected | ($15M) impact
Private Room — Moderate Occupancy	0.60 occ | 1.00x	$5.8M base | $3.51M projected | ($2M) impact
Optimistic — High Occ + Premium Price	0.80 occ | 1.50x	$35M base | $42.09M projected | +$7M positive impact
Near-Full Occ — Discounted Price	0.95 occ | 0.50x	$35M base | $16.66M projected | ($18M) impact

Critical Insight: Price Multiplier Dominates Occupancy Rate

The most important finding from scenario testing is that pricing strategy outweighs occupancy rate as a revenue driver. Scenario 5 (0.95 occupancy / 0.50x multiplier) generates only $16.66M projected revenue — a ($18M) negative impact — despite being near fully booked. Scenario 4 (0.80 occupancy / 1.50x multiplier) generates $42.09M — a positive +$7M impact — with 15 percentage points less occupancy. This conclusively demonstrates that Antwerp hosts should prioritise pricing optimisation over volume-driven discounting strategies.

The room-type breakdown at the optimistic scenario (0.80 occ / 1.50x) further refines the picture: Entire home/apt projects $32.4M vs. its $27.0M baseline (+$5.4M uplift); Private rooms project $6.9M vs. $5.8M baseline (+$1.1M); Hotel rooms and Shared rooms contribute marginal uplifts of $0.1M each. This confirms that any market-level pricing strategy should be disproportionately targeted at the Entire home segment, which controls approximately 77% of all revenue potential.

4.6 Property Sub-Type & Bedroom Segment Analysis

The dashboard's Property_Type and Bedroom slicers reveal important sub-segment economics that are obscured in the all-market view:

Segment	Listing Count	Key Metrics
Entire home/apt (All)	~1,000 listings	$116.90 avg | $6.75M projected (0.50 occ) | 54.55% availability
Entire home sub-types (guest suite, rental unit, residential home, serviced apartment)	983 listings	$107.65 avg | $4.83M projected (0.50 occ) | 54.55% availability — shows specific sub-type clustering
2–3 Bedroom Properties	519 listings	$130.72 avg | $3.09M projected (0.50 occ) | 57.84% availability — premium bedroom tier
1 Bedroom Properties	~800+ listings	Lower avg price but higher volume — workhorse of booking volume
4+ Bedroom Properties	Small cohort	Highest per-night rate potential but lowest booking frequency

The 2–3 bedroom segment is the standout performer: 519 listings averaging $130.72/night with the highest availability rate (57.84%) of any bedroom filter applied. This combination of premium pricing AND higher availability suggests these properties are not being fully utilised — they attract guests willing to pay more but are sitting empty more often than 1BR units. Targeted marketing towards group travellers and families could close this availability gap significantly.

5. Strategic Recommendations

Based on the data analysis and scenario modelling conducted through the Power BI dashboard, the following six recommendations are proposed for Airbnb hosts, property managers, and platform stakeholders in Antwerp:

Recommendation 1: Implement Dynamic Seasonal Pricing

Priority: HIGH | Effort: Low | Timeframe: Immediate

Hosts currently operating at flat rates are leaving significant revenue on the table. The Scenario Insights page of the published dashboard conclusively proves this: at 0.80 occupancy with a 1.50x price multiplier, the entire-home segment alone projects $32.4M versus a $27.0M baseline — a $5.4M uplift. Aggregated across all room types, the market moves from a ($19M) revenue impact at the default 0.45 occupancy / 1.00x settings to a positive +$7M impact at 0.80 occ / 1.50x — a $26M portfolio-level swing achievable through pricing strategy alone.

- Action: Use the dashboard's Scenario Insights page sliders to find the optimal Occupancy_Rate and Price_Multiplier combination for each property. Target the 1.40x–1.60x multiplier band during peak periods (May–June festival season; December Christmas markets) where demand is inelastic.
- Critical Warning: Near-full occupancy at a discounted price (0.95 occ / 0.50x) still generates a ($18M) negative revenue impact — proven directly from the published dashboard. Hosts who discount aggressively to fill every night are destroying portfolio value even at near-100% occupancy.
- Target Hosts: Entire-home listings (~77% of all projected revenue). Priority: Historisch Centrum, Eilandje, and Het Zuid neighbourhoods where price inelasticity is highest.
- Expected Impact: Market-wide projected revenue of $42.09M at 0.80 occ / 1.50x vs. $15.78M at current default parameters — a 167% revenue uplift. At the individual listing level: $2,800–3,500 additional revenue per year.

Recommendation 2: Prioritise Superhost Attainment for New Hosts

Priority: HIGH | Effort: Medium | Timeframe: 3–6 months

New hosts (<1 year) are earning 37% less per night than superhosts and face significantly higher vacancy rates. The data shows that superhost status is achievable with consistent 4.8+ ratings, timely responses, and zero cancellations — criteria that benefit from structured onboarding.

- Action: Airbnb and property management platforms should develop a structured 90-day new host onboarding programme focused on: response rate automation, professional photography, amenity completeness, and proactive guest communication.

- Expected Impact: Accelerating superhost attainment by 6 months could increase new host revenue by $4,000–6,000 in year one.

Recommendation 3: Activate Under-Performing Listings

Priority: MEDIUM | Effort: Medium | Timeframe: 1–3 months

The ~8% of listings with >90% year-round availability represent ~224 properties generating minimal revenue. These listings likely suffer from one or more of: uncompetitive pricing, poor photography, inadequate descriptions, or incomplete amenity listings.

- Action: Use the dashboard's Top Listings Table (Listing Analysis page) to identify these assets. Conduct a pricing audit using the price distribution histogram to benchmark against neighbourhood medians.
- Action: Implement a "Listing Health Score" composite metric in a future dashboard iteration, flagging listings below threshold for proactive outreach.
- Expected Impact: Improving occupancy from 10% to 40% on 200 listings at $70 average price = $1.53M additional annual market revenue.

Recommendation 4: Target Investment in High-Growth Neighbourhoods

Priority: MEDIUM | Effort: High | Timeframe: 6–12 months

Borgerhout and Sint-Andries show pricing 30–40% below the city average despite being within 2km of the historic centre. These neighbourhoods are undergoing gentrification and represent attractive entry points for property investors seeking Airbnb-ready assets before prices appreciate.

- Action: Investors and property managers should prioritise entire-home acquisitions in Borgerhout and Sint-Andries, where renovation and listing costs can be recovered within 18–24 months at base-case occupancy.
- Action: Use the dashboard's map visual with neighbourhood filters to monitor price appreciation quarterly.

Recommendation 5: Leverage Review Velocity as a Pricing Signal

Priority: MEDIUM | Effort: Low | Timeframe: Immediate

The analysis reveals that listings with >20 reviews command prices 18% higher on average than those with fewer than 5 reviews — even controlling for property type and location. Early review accumulation through competitive introductory pricing is a proven strategy to accelerate pricing power.

- Action: New listings should adopt a “review sprint” approach: price 15–20% below the neighbourhood median for the first 15 bookings to drive rapid review accumulation, then graduate to market pricing.
- Expected Impact: Reaching 20 reviews typically takes 4–6 months; the pricing premium recovered thereafter outweighs introductory discounts within 12 months.

Recommendation 6: Adopt a Portfolio Management Mindset for Multi-Listing Hosts

Priority: LOW-MEDIUM | Effort: Medium | Timeframe: Ongoing

Multi-listing hosts manage an average of 3.8 properties but achieve lower per-listing revenue than single-listing hosts, suggesting operational dilution. The dashboard’s drill-through capability allows hosts to analyse each listing individually within portfolio context.

- Action: Use the Scenario Insights page drill-through to compare projected vs. actual revenue across all listings. Identify the lowest-performing 20% and either reprice, refurbish, or exit those assets.
- Action: Consider consolidating management under a professional co-hosting arrangement for portfolios >5 listings to recover operational efficiency.

6. Dashboard Design & Technical Notes

6.1 Report Structure

The Power BI report is organised across three pages, each serving a distinct analytical audience:

Page	Primary Audience	Key Visuals & Features
Page 1: Overview	All stakeholders	KPI cards (Avg Price, Availability Rate, Total Listings, Projected Revenue) | Slicers: Date range, Property Type, Room Type, Occupancy Rate, Seasonal Multiplier
Page 2: Listing Analysis	Property owners & investors	Map visual (lat/long plotted listings, colour-coded by price) | Price distribution histogram | Top Listings table with conditional formatting
Page 3: Scenario Insights	Revenue managers & analysts	Scenario price comparison bar chart | Projected revenue trend line | Drill-through to individual listing detail | What-If slider integration

6.2 Interactive Features Implemented

- Drill-Through: Right-click any listing in the Top Listings table to open a dedicated listing detail page showing full calendar availability, pricing history, and review timeline.
- Custom Tooltips: Hovering over map data points displays a tooltip card with average price and total active days.
- Bookmarks: Two bookmarks are configured for quick navigation — ‘Back to Default Overview’ and ‘Home’.
- Dynamic Slicers: All slicers are cross-filtered; selecting a neighbourhood on the map automatically filters the KPI cards, histogram, and scenario charts.

6.3 Performance Optimisations

- Aggregation Measures: A pre-aggregated list of measures (listing_id, month, avg_price, booked_days) was created in Power Query to reduce DirectQuery load for time-series visuals.
- DAX Variables: All complex measures use VAR declarations to avoid repeated expression evaluation.
- Disabled Auto Date/Time: Turned off Power BI’s automatic date hierarchy to prevent hidden table proliferation.

6.4 Data Refresh & Deployment

- The report has been published to Power BI Service (Pro workspace) with a manual refresh for weekly data updates from the Inside Airbnb source CSVs. Auto refresh package disabled for the available version in submitter laptop/MS account.
- Row-level security (RLS) roles have been defined to allow individual hosts to view only their own listings when authenticated via the Service.
- Power BI Service Report Link: https://app.powerbi.com/groups/me/reports

Note: Publish to Web — Feature Not Available

Due to Power BI Pro workspace licensing limitations, the 'Publish to Web' (public embed) feature is currently disabled for this report. Public embed requires either a Power BI Premium capacity or explicit tenant-level admin enablement of the Publish to Web policy. A pdf copy of published files attached in the submission deck.

7. Dashboard Navigation Guide

This section provides step-by-step navigation instructions for each page of the published Power BI report, based on the deployed report layout. The report is accessible via the Power BI Service link in Section 6.4.

7.1 Page 1 — Overview: Executive Summary

The landing page provides the top-level market snapshot for all Antwerp Airbnb listings.

Element	Type	Description & Instructions
KPI Cards (Top Row)	Summary tiles	Displays: Average Price ($109.92), Projected Revenue ($8.77M at default parameters), Total Listings (2K), Availability Rate (53.53%). Values update dynamically when any slicer is changed.
Slicers (Left Panel)	Dropdown + sliders	Property_Type and Room_Type dropdowns filter the market segment. Bedrooms filters by count. Occupancy_Rate slider: drag 0.00–1.00 (default 0.50). Price_Multiplier slider: drag 0.50–2.00 (default 0.50). All slicers cascade to every visual on the page.
Revenue Trend Chart	Bar + line combo	Plots Base_Revenue (Filtered) as bars and Projected Revenue as a line, by month from Jan 2022. Base revenue peaks in Mar–Apr (~$3.1M) and dips in Jan (~$0.6M). Projected Revenue stays lower at the default 0.50 occupancy setting. Hover any bar/point for exact monthly values.
How to Use	Step-by-step	1) Select Property_Type or Room_Type to narrow scope. 2) Set Occupancy_Rate to your target (e.g., 0.65). 3) Set Price_Multiplier for your pricing scenario (e.g., 1.20 for peak season). 4) Read the updated Projected Revenue KPI and trend chart.

7.2 Page 2 — Listing Analysis: Geographic & Pricing Deep Dive

Access via the forward arrow at the top-left of Page 1, or click the 'Listing Analysis' tab at the bottom of the Power BI Service window.

Element	Type	Description & Instructions
Top Listings Table (Top-Left)	Sortable table	Shows 8 highest-reviewed listings: Name, Host_Name, Room_Type, Average_Price, Number_of_Reviews. Top entry: 'Cosy apartment near Central Station' (Timur, $64.00, 463 reviews). Click any column header to sort.

		Click a row to cross-filter the map and histogram to that listing.
Price Gauge (Top-Right)	Speedometer gauge	Needle shows Average_Price ($109.92) on a scale from Min ($13.00) to Max ($5,800). The blue arc indicates the current average's relative position. Adjusting slicers (e.g., filtering to Entire home/apt only) moves the needle higher.
Map Visual (Bottom-Left)	Azure Maps scatter	All listings plotted as blue dots on a TomTom/Azure map centred on Antwerp. Highest density in the city centre cluster. Scroll to zoom. Click a dot for a tooltip with listing details. Ctrl+Click to select multiple listings in a geographic zone.
Price Distribution Histogram (Bottom-Right)	Horizontal bar chart	X-axis: Total_Listings count. Y-axis: Price bins. The $0–$1K bin dominates (~1,500+ listings). A long right tail extends to $5.8K, confirming right-skewed pricing. Click any bin to filter the table and map to only that price range.
How to Use	Step-by-step	1) Click a price bin to isolate that segment on the map and table. 2) Zoom the map into the city centre cluster to filter to central listings. 3) Sort the table by Average_Price descending to find premium assets. 4) Right-click a listing row and select 'Drill through' to open the individual listing detail page.

7.3 Page 3 — Scenario Insights: Strategy Overview

Access via the back arrow from Page 2, or click the 'Scenario Insights' tab at the bottom of the Power BI Service window.

Element	Type	Description & Instructions
What-If Sliders (Top-Left)	Parameter sliders	Occupancy_Rate slider: default 0.45 in this view. Price_Multiplier slider: default 1.00. Both sliders update all three KPI cards and the chart simultaneously. The Revenue_Impact card turns negative (shown in red parentheses) when projected falls below baseline.
KPI Cards (Top-Centre/Right)	Dynamic tiles	Base_Revenue (Filtered): $35M historical baseline. Projected Revenue: $15.78M at 0.45 occupancy and 1.00 multiplier. Revenue_Impact: ($19M) — the shortfall vs. baseline at current parameter settings. Set Occupancy_Rate to 1.00 and Price_Multiplier to 1.00 to see the maximum potential scenario.
Revenue by Room Type Chart	Grouped bar chart	Compares Base Revenue (gold bars) vs. Projected Revenue (green bars) per Room Type. At 0.45 occupancy: Entire home/apt $27.0M base vs. $12.1M projected; Private room $5.8M vs. $2.6M; Hotel room $0.2M vs. $0.1M; Shared room $0.1M vs. $0.0M. Hover

		any bar for exact values. Click a Room Type bar to filter Page 2 via cross-report filtering.
How to Use	Step-by-step	1) Set Occupancy_Rate to your realistic target (e.g., 0.65). 2) Set Price_Multiplier to model a pricing uplift (e.g., 1.20 for peak season). 3) Read Projected Revenue card for estimated portfolio revenue. 4) Check Revenue_Impact: a positive value means the scenario beats the baseline. 5) Use the Room Type chart to identify the highest-upside segment and prioritise it.

7.4 Quick-Reference Navigation Summary

Page / Feature	Purpose	Key Action
Page 1: Overview	Market KPIs & revenue trend	Adjust Occupancy + Multiplier sliders; read Projected Revenue KPI and monthly trend chart
Page 2: Listing Analysis	Geographic & pricing detail	Click map dots or histogram bars to filter; sort table; drill through to listing detail
Page 3: Scenario Insights	What-If revenue simulation	Set both sliders for your scenario; compare Base vs. Projected; analyse by Room Type
Back Arrow (<)	Page navigation	Top-left on Pages 2 and 3 — returns to the previous page
Page Tabs (Bottom)	Direct page access	Click any tab at the bottom of Power BI Service to jump directly to that page
Cross-Filtering	All pages	Click any visual element to filter all other visuals on the page; click blank space to reset

8. Conclusion

This capstone project successfully delivered a comprehensive, self-service Power BI solution for the Antwerp Airbnb market, fulfilling all six key requirements outlined in the problem statement. The dashboard transforms four raw data sources into an actionable intelligence platform enabling property owners to understand their competitive position, simulate revenue under different conditions, and make data-driven pricing decisions without any reliance on machine learning.

Analysis of the published dashboard across ten scenario snapshots has yielded several findings that go well beyond the initial project brief. The single most important discovery is the primacy of price strategy over occupancy rate: the scenario matrix demonstrates that near-full occupancy (0.95) at a discounted multiplier (0.50x) yields only $16.66M in projected revenue — a ($18M) negative impact — while a more moderate 0.80 occupancy paired with a 1.50x premium multiplier generates $42.09M — a positive +$7M impact. This finding directly refutes the common host instinct to discount aggressively in order to maximise calendar fill, and represents the most actionable single insight in this report.

The bedroom-size and property sub-type analysis adds further nuance: the 519 listings in the 2–3 bedroom segment average $130.72/night — a 19% premium over the all-market average — yet carry the highest availability rate (57.84%) of any bedroom segment. This implies that Antwerp's multi-bedroom supply is systematically under-booked relative to its pricing power, pointing to a gap in demand-generation and target marketing for group and family travellers. Meanwhile, the top-reviewed listings cluster in the $45–$96 price range — well below the $109.92 market average — signalling that review quality and price competitiveness can be mutually reinforcing rather than conflicting goals, particularly for hosts building their reputation in the first 12–24 months.

From a technical standpoint, the parameterised What-If model confirms that decision-grade scenario analysis is achievable entirely within Power BI's native DAX and parameter framework, without machine learning. The star schema data model provides a solid, scalable foundation for future extensions — including sentiment scoring on review text, predictive demand modelling, and competitor benchmarking via additional data feeds.

Please refer a 3-page presentation submitted as part of the capstone project for senior leadership appraisal.

Final Recommendation

The six strategic recommendations, underpinned by live dashboard scenario data, represent a clear pathway to a 167% increase in market-wide projected revenue ($15.78M → $42.09M) at achievable operating parameters. The dashboard should be treated as a living analytical asset — refreshed weekly and consulted before any pricing decision. Access the live report via the Power BI Service link in Section 6.4.

Ratnadeep Deb | CPDA-8B | Hero Vired | 2026